Project: Data Science Skills

This is a **group project** to be completed collaboratively by your entire project team. The goal is to practice effective **virtual teamwork**, a core professional skill for data scientists.

Note: You are required to deliver an approach document the week before the project code base is due, and make a **group presentation at the first meetup after the project code base is due**.

Using data, investigate the question:

> **Which data science skills are most valued?**

Treat this as an **exploratory analysis**. There is no single "correct" answer; your evaluation and justification matter more than any specific ranking.

Project Requirements and Grading Criteria

Your group is responsible for the following:

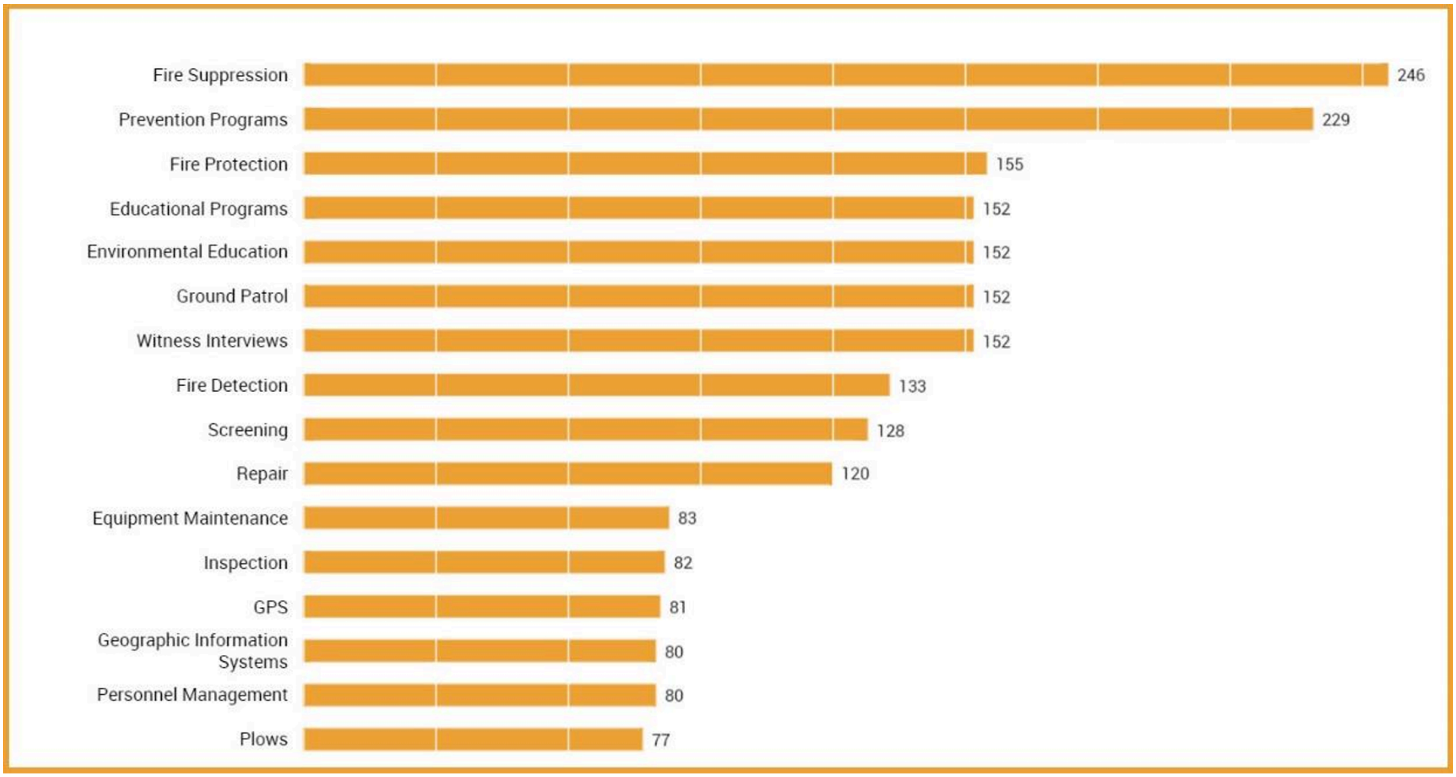
- **Collaboration**
  - Select and use appropriate tools for collaboration (e.g., version control, shared documentation, task tracking).
  - Maintain shared access to code, documentation, and findings.
- **Data Acquisition**
  - Identify relevant data sources. Explain why you chose your data sources.
  - Explain how the data were collected and loaded.
  - Store the data in a **relational database** using a set of **normalized tables**.
- **Data Preparation and Exploration**
  - Perform all data tidying, transformations, and exploratory data analysis in **R**.
  - Address data quality issues such as missing values and outliers.
- **Deliverables**

- Submit all code, results, and documentation.
- Clearly document your **motivation**, **approach**, and **findings**.

- **Presentation**
  - Appoint **at least two group members** to lead different parts of the presentation.
  - All students must be prepared to explain how the data were collected, loaded, transformed, tidied, and examined for issue such as outliers.
- **Statistical Modeling**
  - You are encouraged to experiment with statistical analysis or modeling.
  - **Statistical modeling will not affect your grade**, as this is a first-semester course focused on data acquisition and management.
  - Individual students are **not required** to explain advanced modeling techniques during the presentation.
- **To Consider**
  - Did your team define **operational proxies** for "skill demand" (e.g., job postings, surveys, curricula).
  - Are there comparisons to make across data sources or longitudinally (across time)?
- **Attendance and Participation**
  - If you cannot attend the meetup presentation:
    - Present one-on-one with the instructor **before** the meetup, **or**
    - Submit a **3–7 minute video** explaining your work.

Students are strongly encouraged to **start early**, ask questions, and actively participate in the discussion forum.

Students who are unable or unwilling to work in groups may choose to make Project 3 the one course project that they drop without penalty. There is not an option to deliver Project 3 as a sole contributor.



One *example* approach: The graph above shows Top Forest Ranger Skills, based on number of resumes with specified skills